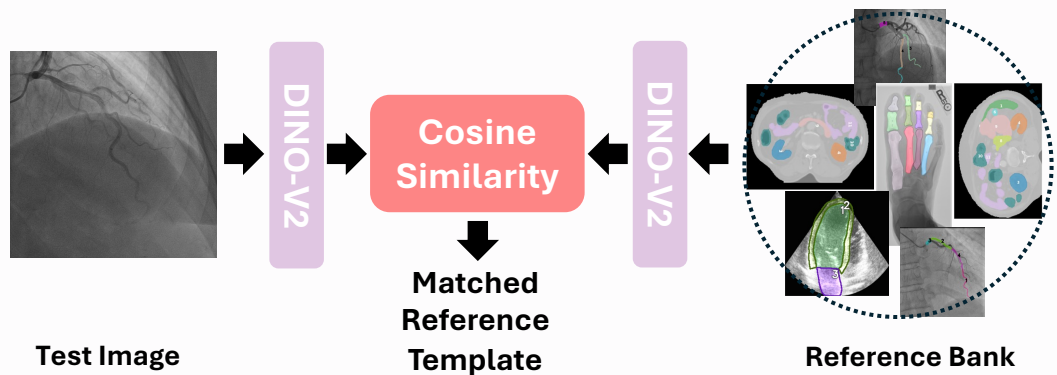


(a) DINO-based Template Matching



(b) Reward Function

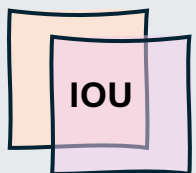
Think Format Reward

<Think> ... <\Think> <Answer> ... <\Answer>

JSON Format Reward

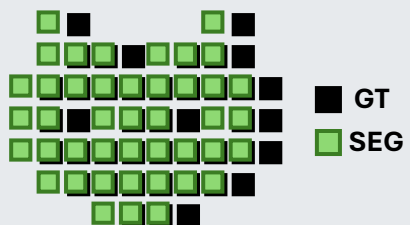
{"Bbox": [20, 25, 232, 142], "CLASS": 3}

Bbox IoU Reward



$> 0.5?$ $\begin{cases} 1 \text{ (YES)} \\ 0 \text{ (NO)} \end{cases}$

Segmentation Reward



$$\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{BCE}} + \beta \cdot \mathcal{L}_{\text{DICE}}$$

(c) Integration with Semantically-Rich Models

An Example Template with 4 Classes

